

Mid BIO 401 Solved

1. What are outlier?

Unusual data values as compared to the rest of the set. They may be distinguished by gaps in a histogram.

2. Name of probability sampling design?

Simple random sampling Stratified sampling Systematic sampling Cluster (area) sampling Multistage sampling

3. Difference between qualitative quantitative variables?

Quantitative Variables It can be measured in the usual sense. For example: - the heights of adult males, - the weights of preschool children, - the ages of patients seen in a - dental clinic.

Qualitative Variables Many characteristics are not capable of being measured. Some of them can be ordered or ranked. For example: - classification of people into socioeconomic groups, -social classes based on income, education, etc

4. What are shapes of distribution?

Symmetric • Data is symmetric if the left half of its histogram is roughly a mirror image of its right half.

Skewed • Data is skewed if it is not symmetric and if it extends more to one side than the other.

Uniform • Data is uniform if it is equally distributed (on a histogram, all the bars are the same height or approximately the same height).

5. What is data and what are difference source of data?

The raw material of Statistics is data. We may define data as figures. Figures result from the process of counting or from taking a measurement. For example: - When a hospital administrator counts the number of patients (counting). - When a nurse weighs a patient (measurement)

Sources:

Routinely kept records.

External sources.

Surveys:

Experiments

6. What is frequency distribution and frequency histogram?

Frequency Distribution • A table that organizes data values into classes or intervals along with number of values that fall in each class (frequency, f). 1. Ungrouped Frequency Distribution – for data sets with few different values. Each value is in its own class. 2. Grouped Frequency Distribution: for data sets with many different values, which are grouped together in the classes.

Frequency Histogram • A bar graph that represents the frequency distribution. • The horizontal scale is quantitative and measures the data values. • The vertical scale measures the frequencies of the classes. • Consecutive bars must touch.